PySpark Interview Coding Questions

1 Read a CSV file into a DataFrame and display the schema.

In [1]:
```
1  from pyspark.sql import SparkSession
2  spark = SparkSession.builder.appName("Q1_Read_CSV").getOrCreate()
3  df = spark.read.option("header", True).option("inferSchema", True)
       .csv("data/employees.csv")
4  df.printSchema()
```

Out[1]:
```
root
 |-- emp_id: integer (nullable = true)
 |-- name: string (nullable = true)
 |-- department: string (nullable = true)
 |-- salary: double (nullable = true)
 |-- join_date: date (nullable = true)
```

2 Filter rows where a column value is greater than a threshold.

In [2]:
```
1  threshold = 60000
2  df_filtered = df.filter(df["salary"] > threshold)
3  df_filtered.show(5, truncate=False)
```

Out[2]:
```
+--------+----------+-------------+---------+------------+
| emp_id | name     | department  | salary  | join_date  |
+--------+----------+-------------+---------+------------+
| 2      | Bob      | Engineering | 75000.0 | 2021-06-15 |
| 4      | David    | Engineering | 90000.0 | 2022-01-20 |
| 5      | Eva      | Marketing   | 65000.0 | 2021-03-10 |
| 7      | Grace    | Engineering | 72000.0 | 2020-11-05 |
| 8      | Henry    | Sales       | 61000.0 | 2021-07-19 |
+--------+----------+-------------+---------+------------+
```

3 Select specific columns from a DataFrame.

In [3]:
```
1  df_selected = df.select("emp_id", "name", "salary")
2  df_selected.show(5, truncate=False)
```

Out[3]:
```
+----------+----------+----------+
| emp_id   | name     | salary   |
+----------+----------+----------+
| 1        | Alice    | 50000.0  |
| 2        | Bob      | 75000.0  |
| 3        | Charlie  | 48000.0  |
| 4        | David    | 90000.0  |
| 5        | Eva      | 65000.0  |
+----------+----------+----------+
```

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4 Rename one or more columns.

In [4]:
```
# Rename single column
df_renamed = df.withColumnRenamed("name", "employee_name") \
               .withColumnRenamed("salary", "annual_salary")
df_renamed.show(5, truncate=False)
```

Out[4]:

emp_id	employee_name	department	annual_salary	join_date
1	Alice	HR	50000.0	2021-01-10
2	Bob	Engineering	75000.0	2021-06-15
3	Charlie	Marketing	48000.0	2021-03-10
4	David	Engineering	90000.0	2022-01-20
5	Eva	Marketing	65000.0	2021-07-18

5 Add a new derived column using withColumn.

In [5]:
```
# Add a new column: bonus = 10% of salary
from pyspark.sql.functions import col
df_with_bonus = df.withColumn("bonus", (col("salary") * 0.10).cast("double"))
df_with_bonus.show(5, truncate=False)
```

Out[5]:

emp_id	name	department	salary	join_date	bonus
1	Alice	HR	50000.0	2021-01-10	5000.0
2	Bob	Engineering	75000.0	2021-06-15	7500.0
3	Charlie	Marketing	48000.0	2021-03-10	4800.0
4	David	Engineering	90000.0	2022-01-20	9000.0
5	Eva	Marketing	65000.0	2021-07-18	6500.0

6 Drop duplicate rows.

In [6]:
```
df_dedup = df.dropDuplicates()
df_dedup.show(5, truncate=False)
```

Out[6]:

emp_id	name	department	salary	join_date
1	Alice	HR	50000.0	2021-01-10
2	Bob	Engineering	75000.0	2021-06-15
3	Charlie	Marketing	48000.0	2021-03-10
4	David	Engineering	90000.0	2022-01-20
5	Eva	Marketing	65000.0	2021-07-18

Note: The above outputs are in table format for better readability.

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7 Count null values in each column.

In [7]:
```
from pyspark.sql.functions import col, sum as _sum, when, lit
null_count_df = df.select([_sum(when(col(c).isNull(), 1).otherwise(0)).alias(c)
                for c in df.columns])
null_count_df.show(truncate=False)
```

Out[7]:
```
+--------+-------+------------+--------+-----------+
| emp_id | name  | department | salary | join_date |
+--------+-------+------------+--------+-----------+
| 0      | 1     | 1          | 1      | 1         |
+--------+-------+------------+--------+-----------+
```

Note: Output shows count of nulls in each column.

8 Replace nulls with default values.

In [8]:
```
from pyspark.sql.functions import lit
df_filled = df.fillna({
    'name': 'Unknown',
    'department': 'Not Assigned',
    'salary': 0.0,
    'join_date': '1970-01-01'
})
df_filled.show(5, truncate=False)
```

Out[8]:
```
+--------+---------+--------------+---------+------------+
| emp_id | name    | department   | salary  | join_date  |
+--------+---------+--------------+---------+------------+
| 1      | Alice   | HR           | 50000.0 | 2021-01-10 |
| 2      | Bob     | Engineering  | 75000.0 | 2021-06-15 |
| 3      | Charlie | Marketing    | 48000.0 | 2021-03-10 |
| 4      | David   | Engineering  | 90000.0 | 2022-01-20 |
| 5      | Eva     | Marketing    | 65000.0 | 2021-07-18 |
+--------+---------+--------------+---------+------------+
```

9 Use groupBy and agg to compute sum, count, average, min, max.

In [9]:
```
from pyspark.sql.functions import sum as _sum, count, avg, min as _min, max as _max
agg_df = df.groupBy("department").agg(
    _sum("salary").alias("total_salary"),
    count("*").alias("count"),
    avg("salary").alias("avg_salary"),
    _min("salary").alias("min_salary"),
    _max("salary").alias("max_salary")
).orderBy("department")
agg_df.show(truncate=False)
```

Out[9]:
```
+-------------+--------------+-------+----------------+------------+------------+
| department  | total_salary | count | avg_salary     | min_salary | max_salary |
+-------------+--------------+-------+----------------+------------+------------+
| Engineering | 165000.0     | 2     | 82500.0        | 75000.0    | 90000.0    |
| HR          | 50000.0      | 1     | 50000.0        | 50000.0    | 50000.0    |
| Marketing   | 113000.0     | 2     | 56500.0        | 48000.0    | 65000.0    |
+-------------+--------------+-------+----------------+------------+------------+
```

Note: Assumes a sample DataFrame 'df' with columns: emp_id, name, department, salary, join_date.

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10 Find the top N records by a metric.

In [10]:
```
N = 3
top_n_df = df.orderBy(col("salary").desc()) \
            .limit(N)
top_n_df.show(truncate=False)
```

Out[10]:
```
+--------+-------+-------------+---------+------------+
| emp_id | name  | department  | salary  | join_date  |
+--------+-------+-------------+---------+------------+
| 4      | David | Engineering | 90000.0 | 2022-01-20 |
| 2      | Bob   | Engineering | 75000.0 | 2021-06-15 |
| 5      | Eva   | Marketing   | 65000.0 | 2021-07-18 |
+--------+-------+-------------+---------+------------+
```

11 Join two DataFrames using inner, left, right, and full joins.

In [11]:
```
# Sample DataFrames
df_emp = spark.createDataFrame(
[(1, "Alice", "HR"), (2, "Bob", "Engineering"), (3, "Charlie", "Marketing"), (4, "David", "Engineering")],
 ["emp_id", "name", "department"])

df_dept = spark.createDataFrame(
 [("HR", "New York"), ("Engineering", "San Francisco"), ("Finance", "Chicago")],
["department", "location"])

# Inner Join
inner_df   = df_emp.join(df_dept, on="department", how="inner")
# Left Join
left_df    = df_emp.join(df_dept, on="department", how="left")
# Right Join
right_df   = df_emp.join(df_dept, on="department", how="right")
# Full Outer Join
full_df    = df_emp.join(df_dept, on="department", how="outer")
```

Out[11]:

Inner Join

```
+------+-------+----------+----------+
|emp_id| name  |department| location |
+------+-------+----------+----------+
| 1    | Alice | HR       | New York |
| 2    | Bob   | Engineering | San Francisco |
| 4    | David | Engineering | San Francisco |
+------+-------+----------+----------+
```

Left Join

```
+------+-------+----------+----------+
|emp_id| name  |department| location |
+------+-------+----------+----------+
| 1    | Alice | HR       | New York |
| 2    | Bob   | Engineering | San Francisco |
| 3    | Charlie | Marketing | null |
| 4    | David | Engineering | San Francisco |
+------+-------+----------+----------+
```

Right Join

```
+------+-------+----------+----------+
|emp_id| name  |department| location |
+------+-------+----------+----------+
| 1    | Alice | HR       | New York |
| 2    | Bob   | Engineering | San Francisco |
| 4    | David | Engineering | San Francisco |
| null | null  | Finance  | Chicago  |
+------+-------+----------+----------+
```

Full Outer Join

```
+------+-------+----------+----------+
|emp_id| name  |department| location |
+------+-------+----------+----------+
| 1    | Alice | HR       | New York |
| 2    | Bob   | Engineering | San Francisco |
| 3    | Charlie | Marketing | null |
| 4    | David | Engineering | San Francisco |
| null | null  | Finance  | Chicago  |
+------+-------+----------+----------+
```

12 Perform a self-join on a DataFrame.

In [12]:
```
# Find employees who are in the same department
df_alias1 = df.alias("e1")
df_alias2 = df.alias("e2")

self_join_df = df_alias1.join(
    df_alias2,
    (col("e1.department") == col("e2.department")) & (col("e1.emp_id") < col("e2.emp_id"))
) \
   .select(  col("e1.emp_id").alias("emp_id_1"),
             col("e1.name").alias("emp_name_1"),
             col("e2.emp_id").alias("emp_id_2"),
             col("e2.name").alias("emp_name_2"),
             col("e1.department") )
self_join_df.show(truncate=False)
```

Out[12]:
```
+-----------+------------+----------+------------+--------------+
| emp_id_1  | emp_name_1 | emp_id_2 | emp_name_2 | department   |
+-----------+------------+----------+------------+--------------+
| 2         | Bob        | 4        | David      | Engineering  |
| 2         | Bob        | 5        | Eva        | Engineering  |
+-----------+------------+----------+------------+--------------+
```

Note: Assumes a sample DataFrame 'df' with columns: emp_id, name, department, salary, join_date.

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13 Use window functions to calculate rank, dense rank, and row number.

In [13]:
```
from pyspark.sql import Window
from pyspark.sql.functions import col, rank, dense_rank, row_number

window_spec = Window.partitionBy("department").orderBy(col("salary").desc())

df_ranked = (
    df.withColumn("rank", rank().over(window_spec))
      .withColumn("dense_rank", dense_rank().over(window_spec))
      .withColumn("row_number", row_number().over(window_spec))
)
df_ranked.show(truncate=False)
```

Out[13]:

emp_id	name	department	salary	rank	dense_rank	row_number
4	David	Engineering	90000.0	1	1	1
2	Bob	Engineering	75000.0	2	2	2
8	Henry	Engineering	75000.0	2	2	3
1	Alice	HR	50000.0	1	1	1
6	Frank	HR	50000.0	1	1	2
5	Eva	Marketing	65000.0	1	1	1
3	Charlie	Marketing	48000.0	2	2	2

Note: rank() gives gaps in ranking for ties. dense_rank() does not give gaps. row_number() is unique sequence.

14 Find the second highest salary.

In [14]:
```
from pyspark.sql.functions import desc
df.select("salary").distinct().orderBy(desc("salary")).limit(1).offset(1).show()
```

Out[14]:

salary
75000.0

Note: distinct() removes duplicates. Order desc, skip first (highest), take next one.

15 Calculate running totals using window functions.

In [15]:
```
from pyspark.sql import Window
from pyspark.sql.functions import sum as _sum

window_spec = Window.partitionBy("department").orderBy(col("salary").desc()) \
              .rowsBetween(Window.unboundedPreceding, Window.currentRow)
df_running =  df.withColumn("running_total_salary", _sum(col("salary")).over(window_spec)) \
              .orderBy("department", desc("salary"))
df_running.show(truncate=False)
```

Out[15]:

emp_id	name	department	salary	running_total_salary
4	David	Engineering	90000.0	90000.0
2	Bob	Engineering	75000.0	165000.0
8	Henry	Engineering	75000.0	240000.0
1	Alice	HR	50000.0	50000.0
6	Frank	HR	50000.0	100000.0
5	Eva	Marketing	65000.0	65000.0
3	Charlie	Marketing	48000.0	113000.0

Note: rowsBetween(Window.unboundedPreceding, Window.currentRow) gives running total up to current row.

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16 Pivot the DataFrame to get total salary by department.

In [16]:
```
from pyspark.sql.functions import sum as _sum
pivot_df = df.groupBy("department").pivot("department").agg(_sum("salary").alias("total_salary"))
pivot_df.fillna(0).orderBy("department").show(truncate=False)
```

Out[16]:
```
+-------------+--------------+
| department  |  total_salary|
+-------------+--------------+
| Engineering |    240000.0  |
| HR          |    100000.0  |
| Marketing   |    113000.0  |
+-------------+--------------+
```

Note: Pivoted on 'department' to calculate total salary for each department.

17 Find employees who earn more than the average salary in their department.

In [17]:
```
from pyspark.sql import Window
from pyspark.sql.functions import avg, col

# Average salary per department
w = Window.partitionBy("department")
df_with_avg = df.withColumn("avg_salary", avg("salary").over(w))

# Filter employees earning more than department average
result_df = df_with_avg.filter(col("salary") > col("avg_salary")) \
            .drop("avg_salary").orderBy("department", "salary", ascending=False)
result_df.show(truncate=False)
```

Out[17]:
```
+---------+----------+-------------+---------+------------+
| emp_id  |  name    | department  |  salary  | join_date  |
+---------+----------+-------------+----------+------------+
| 4       |  David   | Engineering |  90000.0 | 2022-01-20 |
| 2       |  Bob     | Engineering |  75000.0 | 2021-06-15 |
| 5       |  Eva     | Marketing   |  65000.0 | 2021-07-18 |
+---------+----------+-------------+----------+------------+
```

Note: Compares each employee's salary with average salary of their department.

18 Find the month with the highest total salary paid.

In [18]:
```
from pyspark.sql.functions import to_date, date_format, sum as _sum
df_with_month = df.withColumn("month", date_format(to_date("join_date"), "yyyy-MM"))
monthly_salary = df_with_month.groupBy("month").agg(_sum("salary").alias("total_salary"))
monthly_salary.orderBy(desc("total_salary")).limit(1).show(truncate=False)
```

Out[18]:
```
+--------+-------------+
| month  | total_salary|
+--------+-------------+
| 2022-01|    90000.0  |
+--------+-------------+
```

Note: Converts join_date to month (yyyy-MM) and aggregates total salary per month.

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19 Find duplicate records based on all columns.

In [19]:
```
from pyspark.sql.functions import count
dup_df = df.groupBy(df.columns).agg(count("*").alias("count")) \
           .filter("count > 1").drop("count")
dup_df.show(truncate=False)
```

Out[19]:
```
+--------+-------+-------------+---------+-----------+
| emp_id | name  | department  | salary  | join_date |
+--------+-------+-------------+---------+-----------+
| 2      | Bob   | Engineering | 75000.0 | 2021-06-15|
| 5      | Eva   | Marketing   | 65000.0 | 2021-07-18|
+--------+-------+-------------+---------+-----------+
```

Note: Returns rows that appear more than once in the DataFrame.

20 Find the top earning employee in each department.

In [20]:
```
from pyspark.sql import Window
from pyspark.sql.functions import row_number
w = Window.partitionBy("department").orderBy(col("salary").desc())
df_with_rank = df.withColumn("rn", row_number().over(w))
top_emp_per_dept = df_with_rank.filter(col("rn") == 1).drop("rn")
top_emp_per_dept.show(truncate=False)
```

Out[20]:
```
+--------+--------+-------------+---------+------------+
| emp_id | name   | department  | salary  | join_date  |
+--------+--------+-------------+---------+------------+
| 4      | David  | Engineering | 90000.0 | 2022-01-20 |
| 1      | Alice  | HR          | 50000.0 | 2021-01-10 |
| 5      | Eva    | Marketing   | 65000.0 | 2021-07-18 |
+--------+--------+-------------+---------+------------+
```

Note: Uses row_number over window partitioned by department to get top salary record per department.

21 Find employees who have not joined in the given year. (e.g., year = 2021)

In [21]:
```
year = 2021
df_not_joined = df.filter(year(col("join_date")) != year)
df_not_joined.show(truncate=False)
```

Out[21]:
```
+--------+--------+-------------+---------+-----------+
| emp_id | name   | department  | salary  | join_date |
+--------+--------+-------------+---------+-----------+
| 4      | David  | Engineering | 90000.0 | 2022-01-20|
| 7      | Grace  | HR          | 58000.0 | 2020-12-05|
+--------+--------+-------------+---------+-----------+
```

Note: Filters rows where the year part of join_date is not equal to the given year.

22 Find departments that have more than 3 employees.

In [22]:
```
from pyspark.sql.functions import count
df.groupBy("department").agg(count("emp_id").alias("emp_count")) \
          .filter(col("emp_count") > 3) \
          .orderBy("emp_count", ascending=False).show(truncate=False)
```

Out[22]:
```
+--------------+-----------+
| department   | emp_count|
+--------------+-----------+
| Engineering  |     4     |
| Marketing    |     4     |
+--------------+-----------+
```

Note: Returns departments where number of employees is greater than 3.

23 Find the longest employee name.

In [23]:
```
from pyspark.sql.functions import length, desc
df.select("name", length(col("name")).alias("name_length")) \
        .orderBy(desc("name_length")).limit(1).show(truncate=False)
```

Out[23]:
```
+-------------+------------+
| name        | name_length|
+-------------+------------+
| Christopher |     11     |
+-------------+------------+
```

Note: Calculates length of each name and returns the longest one.

24 Add a new column "year_of_joining" from "join_date".

In [24]:
```
from pyspark.sql.functions import year
df_with_year = df.withColumn("year_of_joining", year(col("join_date")))
df_with_year.show(truncate=False)
```

Out[24]:
```
+--------+-------+-------------+---------+------------+----------------+
| emp_id | name  | department  | salary  | join_date  | year_of_joining|
+--------+-------+-------------+---------+------------+----------------+
| 4      | David | Engineering | 90000.0 | 2022-01-20 |      2022      |
| 2      |  Bob  | Engineering | 75000.0 | 2021-06-15 |      2021      |
| 5      |  Eva  | Marketing   | 65000.0 | 2021-07-18 |      2021      |
| 1      | Alice | HR          | 50000.0 | 2021-01-10 |      2021      |
+--------+-------+-------------+---------+------------+----------------+
```

Note: Extracts year from join_date and adds as a new column.

25 Find pairs of employees who are in the same department.

In [25]:
```
e1 = df.alias("e1")
e2 = df.alias("e2")
pairs_df = e1.join(e2, (col("e1.department") == col("e2.department")) & (col("e1.emp_id") < col("e2.emp_id")), "inner") \
        .select(col("e1.department").alias("department"), col("e1.name").alias("emp1"), col("e2.name").alias("emp2")) \
        .orderBy("department", "emp1", "emp2")
pairs_df.show(truncate=False)
```

Out[25]:
```
+-------------+----------+--------+
| department  |  emp1    |  emp2  |
+-------------+----------+--------+
| Engineering |  Bob     |  David |
| HR          |  Alice   |  Grace |
| Marketing   |  Charlie |  Eva   |
+-------------+----------+--------+
```

Note: Self-join on department and keep pairs where emp_id of e1 is less than e2 to avoid duplicates and self-pairs.